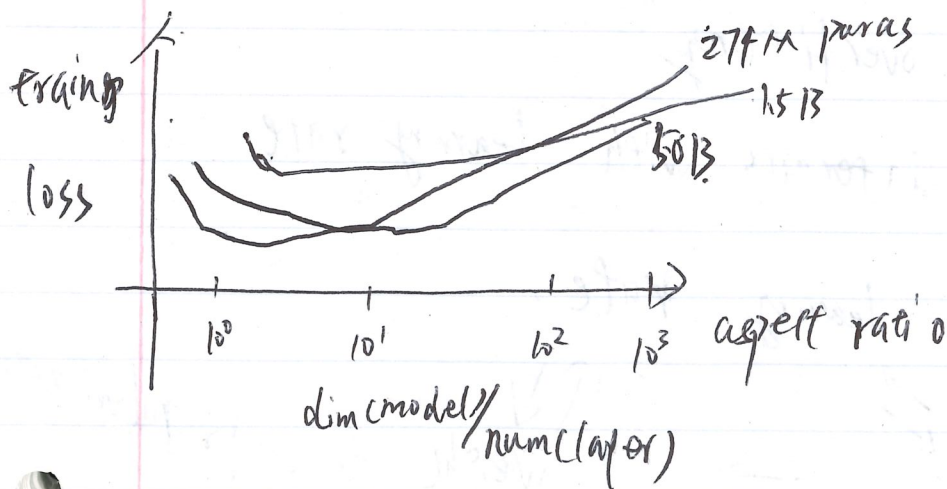
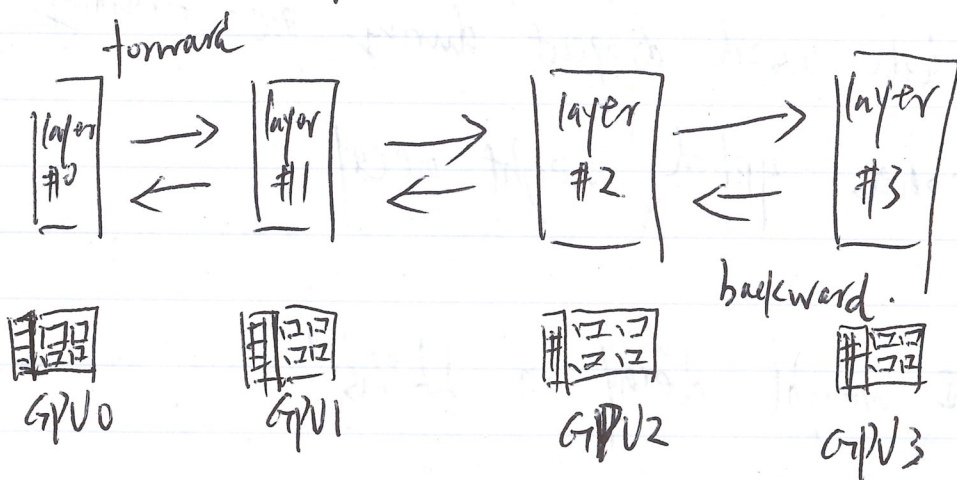


should my model be deep or wide? How deep and how wide?

$\text{dim(model)/nlayer} = \text{Sweet spot}$

↓ Extremely deep models are harder to parallelize and have higher latency

↓ scaling limitation: Depth vs Width.



What are typical vocabulary size?

30-50k tokens $\Rightarrow$ Monolingual : GPT 2/3

100-250k tokens $\Rightarrow$ Multilingual : GPT 5 / deepseek

Dropout / regularization = we need regularization during pre-training.

$\downarrow$
old models used "dropout" during pre-training
new models applied "weight decay"

$\downarrow$
why ~~do~~ weight decay in LLMs?

not to control overfitting

weight decay interacts with learning rate.

